Problems

5. The following is a frequency table of daily travel times (in minutes):
   a. How many days are reported in the frequency table?
   b. Find the sum of the travel times of all those days.

Travel time	Frequency
15	6
18	5
22	4
23	3
24	4
25	2
26	4
32	3
36	1
48	1

6. Table 2.9 gives the number of commercial airline accidents and the total number of resulting fatalities in the United States in the years from 1985 to 2006.
   a. Represent the number of yearly airline accidents in a frequency table.
   b. Give a frequency polygon graph of the number of yearly airline accidents.
   c. Give a cumulative relative frequency plot of the number of yearly airline accidents.
   d. Find the sample mean of the number of yearly airline accidents.
   e. Find the sample median of the number of yearly airline accidents.
   f. Find the sample mode of the number of yearly airline accidents.
   g. Find the sample standard deviation of the number of yearly airline accidents.

Table 2.9 U.S. Airline Safety, Scheduled Commercial Carriers, 1985–2006.

Year	Departures (millions)	Accidents	Fatalities	Year	Departures (millions)	Accidents	Fatalities
1985	6.1	4	197	1996	7.9	3	342
1986	6.4	2	5	1997	9.9	3	3
1987	6.6	4	231	1998	10.5	1	1
1988	6.7	3	285	1999	10.9	2	12
1989	6.6	11	278	2000	11.1	2	89
1990	7.8	6	39	2001	10.6	6	531
1991	7.5	4	62	2002	10.3	0	0
1992	7.5	4	33	2003	10.2	2	22
1993	7.7	1	1	2004	10.8	1	13
1994	7.8	4	239	2005	10.9	3	22
1995	8.1	2	166	2006	11.2	2	50

Source: National Transportation Safety Board.

8. The sample mean of the weights of the adult women of town A is larger than the sample mean of the weights of the adult women of town B. Moreover, the sample mean of the weights of the adult men of town A is larger than the sample mean of the weights of the adult men of town B. Can we conclude that the sample mean of the weights of the adults of town A is larger than the sample mean of the weights of the adults of town B? Explain your answer.

10. A total of 100 people work at company A, whereas a total of 110 work at company B. Suppose the total employee payroll is larger at company A than at company B.
   a. What does this imply about the median of the salaries at company A with regard to the median of the salaries at company B?
   b. What does this imply about the average of the salaries at company A with regard to the average of the salaries at company B?

11. The sample mean of the initial 99 values of a data set consisting of 198 values is equal to 120, whereas the sample mean of the final 99 values is equal to 100. What can you conclude about the sample mean of the entire data set
   a. Repeat when "sample mean" is replaced by "sample median."
   b. Repeat when "sample mean" is replaced by "sample mode."

12. The following table gives the number of pedestrians, classified according to age group and sex, killed in fatal road accidents in England in 1922.
   a. Approximate the sample means of the ages of the males.
   b. Approximate the sample means of the ages of the females.
   c. Approximate the quartiles of the males killed.
   d. Approximate the quartiles of the females killed.

Age	Number of Males	Number of Females
0–5	120	67
5–10	184	120
10–15	44	22
15–20	24	15
20–30	23	25
30–40	50	22
40–50	60	40
50–60	102	76
60–70	167	104
70–80	150	90
80–100	49	27

13. The following are the percentages of ash content in 12 samples of coal found in close proximity:

9.2, 14.1, 9.8, 12.4, 16.0, 12.6, 22.7, 18.9, 21.0, 14.5, 20.4, 16.9

Find the
   a. sample mean, and
   b. sample standard deviation of these percentages.
14. The sample mean and sample variance of five data values are, respectively, $\bar{x} = 104$ and $s^2 = 16$. If three of the data values are 102, 100, 105, what are the other two data values?
15. Suppose you are given the average pay of all working people in each of the 50 states of the United States.
   a. Do you think that the sample mean of the averages for the 50 states will equal the value given for the entire United States?
   b. If the answer to part (a) is no, explain what other information aside from just the 50 averages would be needed to determine the sample mean salary for the entire country. Also, explain how you would use the additional information to compute this quantity.
16. The following data represent the lifetimes (in hours) of a sample of 40 transistors:

112, 121, 126, 108, 141, 104, 136, 134
121, 118, 143, 116, 108, 122, 127, 140
113, 117, 126, 130, 134, 120, 131, 133
118, 125, 151, 147, 137, 140, 132, 119
110, 124, 132, 152, 135, 130, 136, 128

   a. Determine the sample mean, median, and mode.

18. A computationally efficient way to compute the sample mean and sample variance of the data set $x_1, x_2, \ldots, x_n$ is as follows. Let

$$\bar{x}_j = \frac{\sum_{i=1}^{j} x_i}{j}, \quad j = 1, \ldots, n$$

be the sample mean of the first j data values, and let

$$s_j^2 = \frac{\sum_{i=1}^{j} (x_i - \bar{x}_j)^2}{j-1}, \qquad j = 2, \ldots, n$$

be the sample variance of the first j, $j \geq 2$, values. Then, with $s_1^2 = 0$, it can be shown that

$$\bar{x}_{j+1} = \bar{x}_j + \frac{x_{j+1} - \bar{x}_j}{j+1}$$

and

$$s_{j+1}^2 = \left(1 - \frac{1}{j}\right) s_j^2 + (j+1)(\bar{x}_{j+1} - \bar{x}_j)^2$$

a. Use the preceding formulas to compute the sample mean and sample variance of the data values 3, 4, 7, 2, 9, 6.
b. Verify your results in part (a) by computing as usual.
c. Verify the formula given above for $\bar{x}_{j+1}$ in terms of $\bar{x}_j$.

20. Find the quartiles of the following ages at death as given in obituaries of the *New York Times* in the 2 weeks preceding 1 August 2013.

92, 90, 92, 74, 69, 80, 94, 98, 65, 96, 84, 69, 86, 91, 88

74, 97, 85, 88, 68, 77, 94, 88, 65, 76, 75, 60

69, 97, 92, 85, 70, 80, 93, 91, 68, 82, 78, 89

21. The universities having the largest number of months in which they ranked in the top 10 for the number of Google searches over the past 114 months (as of June 2013) are as follows.

University	Number of Months in Top 10
Harvard University	114
University of Texas, Austin	114
University of Michigan	114
Stanford University	113
University of California Los Angeles (UCLA)	111
University of California Berkeley	97
Penn State University	94
Massachusetts Institute of Technology (MIT)	66
University of Southern California (USC)	63
Ohio State University	52
Yale University	48
University of Washington	33

a. Find the sample mean of the data.
b. Find the sample variance of the data.
c. Find the sample quartiles of the data.

23. Represent the data of Problem 20 in a box plot.

30. The following are the heights and starting salaries of 12 law school classmates whose law school examination scores were roughly the same.

Height	Salary
64	91
65	94
66	88
67	103
69	77
70	96
72	105
72	88
74	122
74	102
75	90
76	114

a. Represent these data in a scatter diagram.
b. Find the sample correlation coefficient.